

Learning Roles from Situated Peer Judgments in Multi-Agent Collaboration

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Abstract

Multi-agent collaboration requires decisions about who should do what, yet a team’s final outcome does not identify which contributions helped the agents that depended on them. We study whether judgments made by actual consumers of delivered work can support role formation across tasks among agents without assigned expert roles. We define a role as a context-dependent pattern of responsibilities supported by an agent’s collaboration history. The proposed approach links consumer acceptance, actual use, and repair effort to the producing agent, with later attributable outcomes correcting mistaken judgments. Subsequent collaborators would use this evidence before assigning new work, including collaborators who did not directly evaluate the producer. This makes transfer across collaborators part of the hypothesis. A small decision model updated from delayed feedback on selected peers is a candidate implementation; its backbone and update rule remain to be determined. The planned evaluation uses tasks with genuine dependencies, independent outcome checks, and matched observation and computation budgets. Comparisons with contextual trust, bandit selection, and peer-profile adaptation would test whether situated judgments improve future team quality and total cost. Removing judgments, shuffling their attribution, and withholding their transfer would test the proposed mechanism. The central question is when collaboration history supports useful divisions of responsibility beyond a private preference for the best peer.

CCS Concepts

• Computing methodologies → Multi-agent systems.

Keywords

Multiagent coordination, role learning, peer judgment, online adaptation

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Internal pre-results proposal — not for submission. This document fixes a working question and prospective tests. It does not report an evaluated learning method. The benchmark, backbone, update mechanism, and empirical conclusions remain open.



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1 The Contribution We Need to Establish

The target is a mechanism that uses situated judgments to change useful future responsibilities. It must specify who may observe and share an evaluation, what can be updated after delayed evidence arrives, and how the resulting state changes a decision before work begins. A role label by itself is not an outcome. The intended evidence is a change in actual assignments that improves subsequent quality and complete collaboration cost.

The first development task separates a producer’s code delivery from a recipient’s dependent integration work. These are temporary responsibilities, not preset expert identities. A short, single-handoff probe can establish an executable information boundary; it cannot establish decentralized role formation or the full recursive collaboration setting.

2 Prospective Discriminating Tests

Task and signal qualification. Use real delivered work and record the recipient’s initial judgment, actual use, revisions, and completion cost. Keep independent evaluation outside the agent observation boundary. Distinguish the quality of a delivery from the quality obtained after another agent repairs it. Split by task structure rather than treating renamed instances as independent problems.

Future responsibility. Evaluate legal pre-execution assignments on unseen tasks, including decisions by collaborators who did not directly judge the candidate. Match initial agents, observations, exploration opportunities, memory, and total resources. Give the contextual trust or bandit comparator the same evidence. Test whether task-dependent divisions of responsibility explain any benefit beyond selecting one globally stronger peer.

Mechanism and cost. Remove judgment information, scramble its source within comparable contexts, or withhold its transfer while preserving the remaining execution rules. If real-time training becomes the contribution, compare one explicit updater with online SGD, RLS, and periodic refitting. Measure old-task retention, adaptation, and feedback-to-serving latency jointly; inexpensive parameter updates alone do not establish useful real-time learning.

3 Current Evidence Boundary

The existing executable fixtures test protocol ordering and artifact lineage. Their judgments and repairs are programmed, so they do not support learning efficacy. Earlier API probes establish narrower transport and interaction observations. There is no completed controlled result for the method proposed here. All result claims and a final algorithm description must be supplied from auditable experiments before this proposal can become a submission manuscript.